

- 8 (a) Figure 8.1 shows three processes that require energy in living organisms.

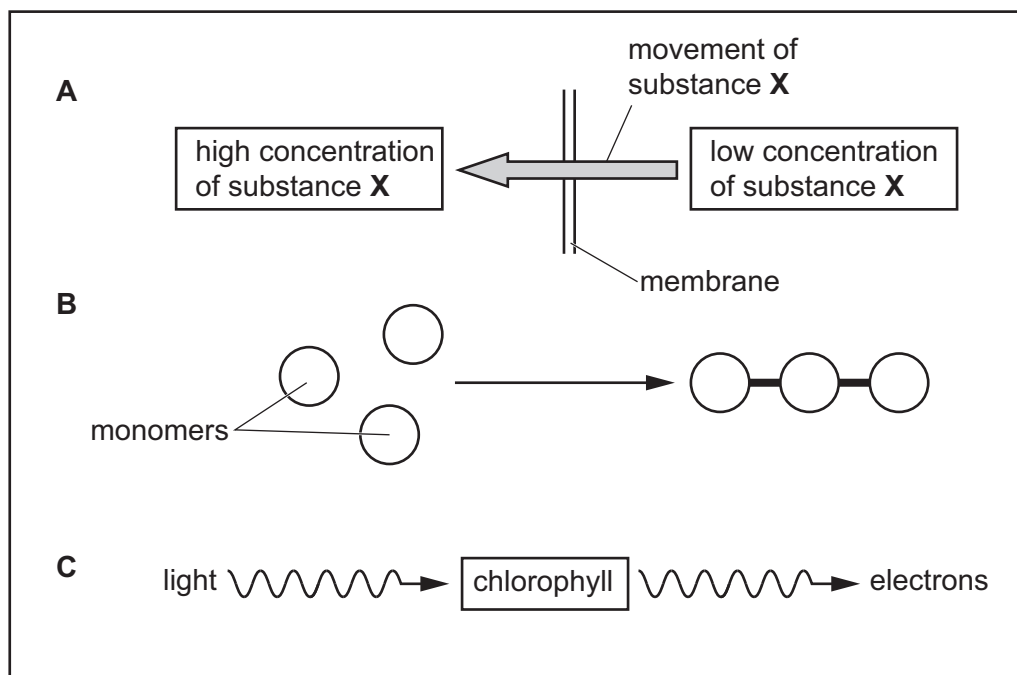


Figure 8.1

State the name of each type of process shown in **A**, **B** and **C**.

A
B
C

[3]

- (b) ATP is produced in mitochondria by a process called chemiosmosis.

Outline the process of chemiosmosis in mitochondria.

.....
.....
.....
.....
..... [2]

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

(c) Different respiratory substrates have different energy values.

Table 8.1 shows the approximate energy values of carbohydrate and lipid.

Table 8.1

respiratory substrate	approximate energy value /kJ g ⁻¹
carbohydrate	17.0
lipid	37.8

Calculate the mass of carbohydrate needed to release the same quantity of energy as 100 g of lipid.

Give your answer to one decimal place.

mass =g [2]

[Total: 7]